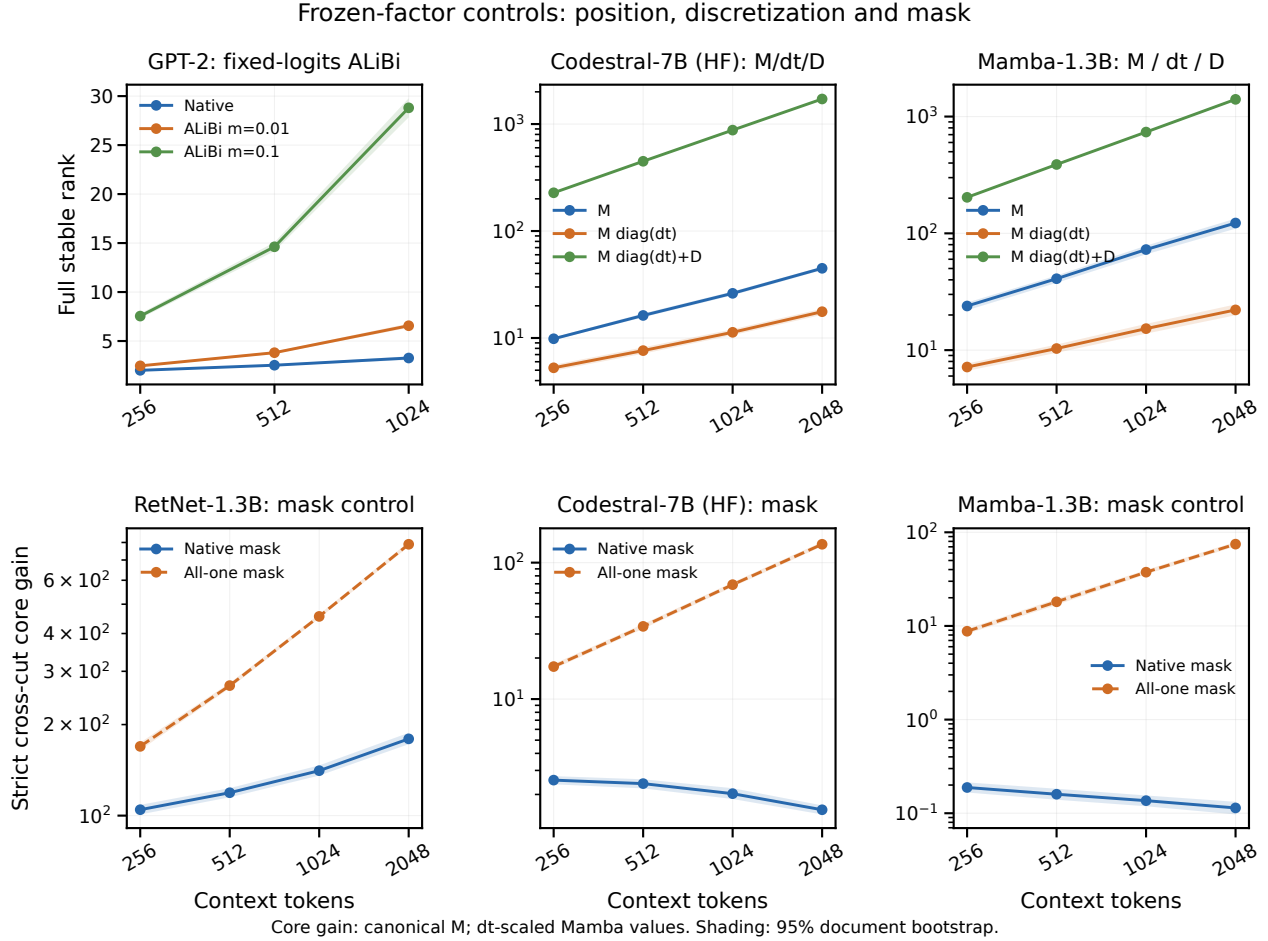
Frozen-factor controls: position, discretization and mask
GPT-2: fixed-logits ALiBi
Codestral-7B (HF): M/dt/D
Mamba-1.3B: M / dt / D
Full stable rank
Native
ALiBi m=0.01
ALiBi m=0.1
M
M diag(dt)
M diag(dt)+D
RetNet-1.3B: mask control
Codestral-7B (HF): mask
Mamba-1.3B: mask control
Strict cross-cut core gain
Native mask
All-one mask
Context tokens
Core gain: canonical M; dt-scaled Mamba values. Shading: 95% document bootstrap.